Page a day worksheet **PAD nr: 1**
Grade 10 – Graphs and Equations – 1

Name:	Memo

QUESTION 1
Solve for x in the following equations. Show ALL calculations.

(a) $x^2 - 3x = 4$ (2)

$x^2 - 3x - 4 = 0$

$(x-4)(x+1) = 0$

Either $x-4=0$ or $x+1=0$

$x = 4$ $x = -1$

(b) $3^{x-1} - 81 = 0$ (3)

$3^{x-1} = 81$

$3^{x-1} = 3^4$

$x-1 = 4$

$x = 5$

[5]

QUESTION 2
Given: $f(x) = 3x^2 - 27$

(a) Calculate the intercepts with the axes. (3)

x-int ($y=0$)

$0 = 3x^2 - 27$

$0 = 3(x^2 - 9)$

$0 = 3(x+3)(x-3)$

$x = -3$ or $x = 3$

y-int ($x=0$)

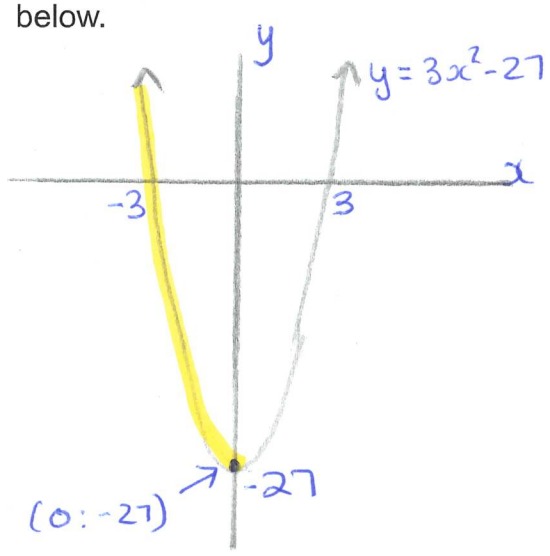
$y = 3(0)^2 - 27$

$y = -27$

(b) Give the range of the graph. (2)

$y \in [-27; \infty)$

(c) Make a neat sketch of the graph below. (4)



(d) For which values of x will the graph be decreasing? (2)

as y decreases, x increases

$\therefore x \in (-\infty; 0)$

[11]

Page a day worksheet **PAD nr: 2**

Grade 10 – Graphs and Equations – 2

Name:	Memo

QUESTION 1

Solve for x in the following equations. Show ALL calculations.

(a) $\left(\frac{1}{2}\right)^x = 64$ (2)

$(2^{-1})^x = 2^6$

$2^{-x} = 2^6$

$-x = 6$

$x = -6$

(c) $(x-2)(x+3) = 6$ (3)

$x^2 + 3x - 2x - 6 - 6 = 0$

$x^2 + x - 12 = 0$

$(x+4)(x-3) = 0$

$x+4 = 0$ or $x-3 = 0$

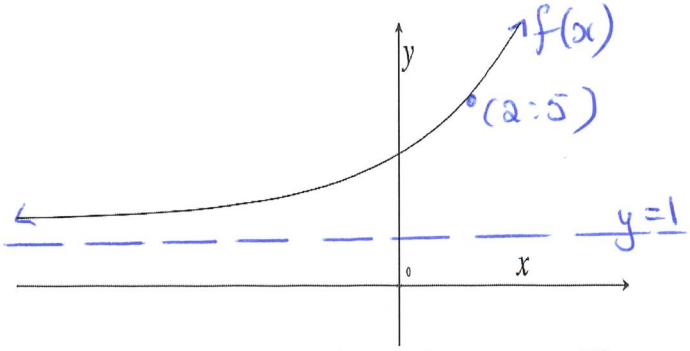
$x = -4$ $x = 3$

[5]

QUESTION 2

The graph of $f(x) = a^x + q$ is drawn:

The graph has an asymptote at $y = 1$ and there is a set of coordinates (2;5) on the graph. Answer the questions below:



(a) Give the range of the graph. (2)

$y \in (1; \infty)$

(b) Determine the values of a and q. (3)

$y = a^x + 1$ $q = 1$

sub (2;5)

$5 = a^2 + 1$

$4 = a^2$

$\pm 2 = a$

$\therefore 2 = a$

$y = 2^x + 1$

(c) Give the equation reflection of the graph in the x-axis (2)

$-y = 2^x + 1$

$\therefore y = -2^x - 1$

(d) How must the graph be translated to be $y = 2^x - 3$? (2)

graph shifts 4 units downwards

[11]

Page a day worksheet **PAD nr: 3**

Grade 10 – Graphs and Equations – 3

Name:	Memo

QUESTION 1

Solve for x in the following equations. Show ALL calculations.

(a) $\frac{3}{x} = \frac{x-1}{2}$ (4)

LCD = $2x$

restriction $x \neq 0$

$\frac{6}{2x} = \frac{x(x-1)}{2x}$

$6 = x^2 - x$

$0 = x^2 - x - 6$

$0 = (x-3)(x+2)$

$x = 3$ or $x = -2$

(c) $25^x - \frac{1}{125} = 0$ (2)

$(5^2)^x = \frac{1}{125}$

$5^{2x} = 125^{-1}$

$5^{2x} = (5^3)^{-1}$

$5^{2x} = 5^{-3}$

$2x = -3$

$x = -3/2$

[6]

QUESTION 2

You are given the function of $f(x) = \frac{-3}{x} + 2$. Answer the following questions:

(a) Give the equations of the asymptotes of f. (2)

$x = 0$ Vertical asymptote

$y = 2$ Horizontal asymptote

(b) Determine the x-intercept of f. (2)

Put $y = 0$

$0 = -\frac{3}{x} + 2$

$-2 = -\frac{3}{x}$

$-2x = -3$

$x = 3/2$ $(\frac{3}{2}; 0)$

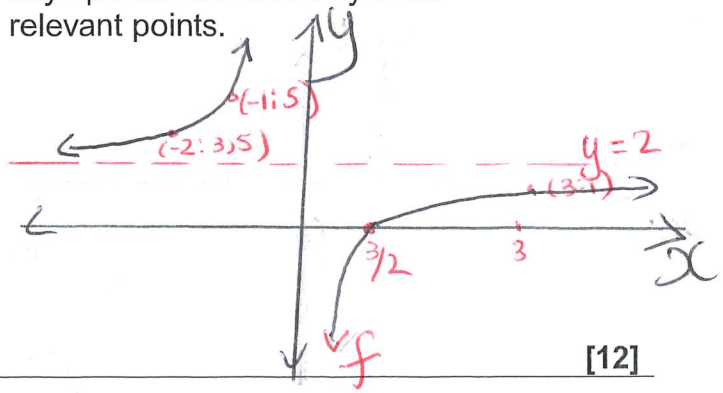
(c) If the coordinates $(-1; a)$, $(-2; b)$ and $(3; c)$ lie on f, determine the values of a, b and c. (3)

$f(-1) = \frac{-3}{-1} + 2 = 5 = a$

$f(-2) = \frac{-3}{-2} + 2 = 3\frac{1}{2} = b$

$f(3) = \frac{-3}{3} + 2 = 1 = c$

(d) Draw a sketch of f and clearly indicate all intercepts with the axes, asymptotes and also any other relevant points. (4)



[12]